Z. 2026 **Concurrency Theory** P. Hofman

Tutorial 4, Friday, 30/10/2026, by vrm

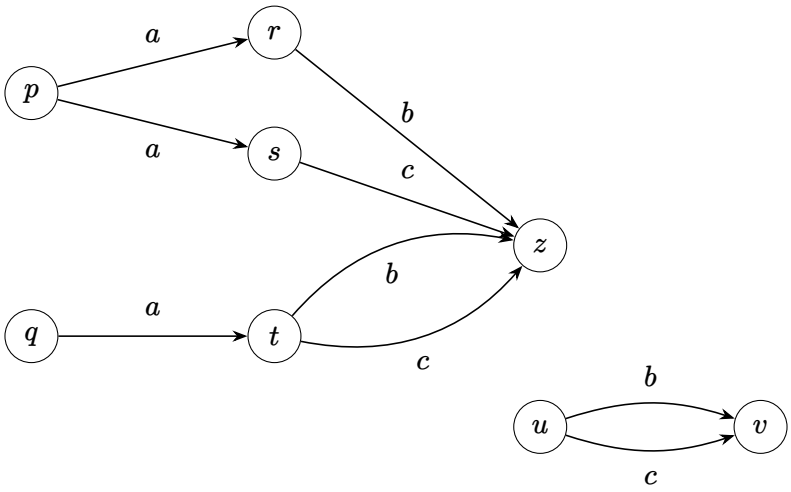
Today: *Bisimulation*

Conventions. All transition systems in this sheet use strong transitions. For a labelled transition system, $\sim$ denotes strong bisimilarity. A relation R is a bisimulation if every move from either side can be matched by a move with the same label, leading again to a pair in R.

For vectors we use componentwise addition and the componentwise order on $\mathbb{N}^k$. A set is *semilinear* if it is a finite union of linear sets.

A. A finite warm-up

Exercise 1. [*Compute bisimilarity*] Consider the following finite LTS.



(a) Compute all bisimulation equivalence classes.

(b) Decide whether $p \sim q$.

(c) Give a winning first move for Spoiler from (p, q).

(d) Give an HML formula which is true in exactly one of p, q.

(e) Compute the first few approximants $\sim_0, \sim_1, \ldots$ until the pair (p, q) disappears.

B. Congruences on $\mathbb{N}^k$

Exercise 2. [*Congruences are semilinear*] An equivalence relation $\equiv$ on $\mathbb{N}^k$ is a *congruence* if

$$x \equiv y \quad \Longrightarrow \quad x + z \equiv y + z \qquad \text{for every } z \in \mathbb{N}^k.$$

We identify $\equiv$ with the set

$$R_{\equiv} = \{(x, y) \in \mathbb{N}^k \times \mathbb{N}^k : x \equiv y\} \subseteq \mathbb{N}^{2k}.$$

A set $S \subseteq \mathbb{N}^d$ is called a *slice* if

$$u,\ u+v,\ u+w \in S \quad \Longrightarrow \quad u+v+w \in S.$$

You may use the following theorem of Eilenberg–Schützenberger:
Every slice $S \subseteq \mathbb{N}^d$ is semilinear.

(a) Let $\equiv$ be a congruence on $\mathbb{N}^k$. Suppose

$$(x,y), \quad (x+p, y+q), \quad (x+r, y+s)$$

belong to $R_{\equiv}$. Prove that

$$(x+p+r, y+q+s) \in R_{\equiv}.$$

(b) Conclude that every congruence on $\mathbb{N}^k$ is a semilinear relation.

Exercise 3. [*Optional challenge: slices are semilinear*] Let $S \subseteq \mathbb{N}^d$ be a slice. For $u \in S$ define

$$S - u = \{v \in \mathbb{N}^d : u+v \in S\}, \qquad M_u = \big(\min((S-u) \setminus \{0\})\big)^*,$$

where $\min X$ denotes the set of componentwise-minimal elements of X, and A^* is the submonoid generated by A.

(a) Use Dickson's lemma to show that every M_u is finitely generated.

(b) Show that $u + M_u \subseteq S$.

(c) Show that if $u \leq v$ and $M_u = M_v$, then

$$v + M_v \subseteq u + M_u.$$

(d) Complete the argument that only finitely many sets $u + M_u$ are needed to cover S.

C. BPP: bisimilarity as a congruence

A BPP has variables $X_1, \ldots, X_k$ and finitely many rules

$$X_i \xrightarrow{a} \alpha, \qquad \alpha \in \mathbb{N}^k.$$

A configuration is a vector $x \in \mathbb{N}^k$. A rule above induces

$$x \xrightarrow{a} x - e_i + \alpha$$

whenever $x_i > 0$ and e_i is a unit vector with 1 on the component i.

Exercise 4. [*Adding the same process*] Prove that

$$x \sim y \quad \Longrightarrow \quad x + z \sim y + z$$

for all BPP configurations x, y, z.

(a) Prove that

$$x \sim y \quad \Longrightarrow \quad x + z \sim y + z$$

for all BPP configurations x, y, z.

(b) Using Exercise 2, conclude that BPP bisimilarity is semilinear.

Exercise 5. [*A symbolic infinite bisimulation*] Consider the BPP with variables X, Y, A and rules

$$\begin{array}{lll} X \xrightarrow{a} X + A, & \qquad & Y \xrightarrow{a} Y + A, \\ X \xrightarrow{b} 0, & & Y \xrightarrow{b} 0, \\ A \xrightarrow{c} 0. & & \end{array}$$

Here $+$ denotes multiset union and 0 is the empty configuration.

Prove that

$$X + nA \sim Y + nA \qquad \text{for every } n \in \mathbb{N}.$$

Give one finite description of an infinite bisimulation witnessing all these pairs at once.

D. Two semi-decision procedures

For an arbitrary LTS define relations $\sim_n$ by

$$\sim_0 = S \times S,$$

and, for $n \geq 0$, let $p \sim_{n+1} q$ if every transition from either state can be matched by a transition with the same label whose endpoints are related by $\sim_n$.

Exercise 6. [*Searching for a negative witness*] Assume that the LTS is finitely branching.

(a) Show that, for fixed p, q, n, the property $p \sim_n q$ can be decided by exploring a finite game tree of depth n.

(b) Prove

$$p \sim q \quad \Longrightarrow \quad p \sim_n q \qquad \text{for every } n.$$

(c) Prove the converse:

$$(\forall n \;\; p \sim_n q) \quad \Longrightarrow \quad p \sim q.$$

Use König's lemma on the tree of finite Duplicator strategies.

(d) Deduce a semi-decision procedure for $p \not\sim q$.

Exercise 7. [*Searching for a positive witness in BPP*] Let x, y be configurations of a fixed BPP.

A semilinear relation $R \subseteq \mathbb{N}^k \times \mathbb{N}^k$ can be represented by a Presburger formula. You may use that Presburger arithmetic is decidable and that semilinear relations can be effectively enumerated.

(a) Write the one-step condition saying that a relation R is a bisimulation.

(b) Design a semi-decision procedure for $x \sim y$, use the fact that semi-linear sets are enumerable.

Remark. Run the procedures from Exercises 5 and 6 in parallel. Note that one has to accept thus strong bisimilarity for BPP is decidable.

E. Why finite branching matters

Exercise 8. [*All finite approximants, but not bisimilar*] For $m \in \mathbb{N}$, let C_m be a finite b-chain of length m:

$$c_m \xrightarrow{b} c_{m-1} \xrightarrow{b} \cdots \xrightarrow{b} c_0,$$

where c_0 has no outgoing transition. Let C_∞ be the infinite b-chain.

Construct states p, q as follows:

$$p \xrightarrow{a} C_m \quad \text{for every } m \in \mathbb{N},$$

and

$$q \xrightarrow{a} C_m \quad \text{for every } m \in \mathbb{N}, \qquad q \xrightarrow{a} C_\infty.$$

(a) Prove that $p \sim_n q$ for every $n \in \mathbb{N}$.

(b) Prove that $p \not\sim q$.

(c) Identify precisely where the proof of Exercise 5(c) fails.

▷ More material here next time. ◁